

Lucas de Almeida Carvalho

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PROFILE

I build and ship production AI systems end-to-end, with a strong focus on LLMs, RAG, and multi-agent architectures. I've built knowledge pipelines covering document and clinical-data ingestion, processing, chunking, embeddings, vector storage, retrieval, and context construction, decoupled from agent runtimes so both can scale independently. I've also built tool-calling agents, LLM evaluation frameworks, and centralized LLM infrastructure for multi-provider model configuration and routing, continuously optimizing for quality, latency, and inference cost.

My background is in Computer Vision, where I've trained and deployed YOLO-based models for real-time object detection, multi-object tracking, and behavioral event recognition across cloud and edge environments, including Apple Silicon. I've shipped multi-camera pipelines achieving 92% precision at real-world confidence thresholds, replacing six-figure SaaS licensing costs with in-house infrastructure.

On the engineering side, I own the full ML lifecycle: data ingestion, model fine-tuning, RAG and retrieval systems, Re-ID, containerized GPU inference, cloud orchestration, and production deployment. I typically operate with end-to-end ownership, making architectural decisions and balancing technical trade-offs from 0 to production.

Experienced working remotely with international teams and communicating technical decisions to both technical and non-technical stakeholders. Async-friendly, English communication.

TECHNICAL SKILLS

- **LLMs & Generative AI:** Multi-agent Architectures, Tool Calling / Agentic Workflows, RAG Pipelines, STT (Speech-to-Text), Prompt Engineering, Structured Reasoning, Structured Output Validation, Model Evaluation & Traceability
- **Primary Language:** Python (Production-grade ML & Backend systems), FastAPI
- **Computer Vision & Edge AI:** YOLO (v8+), Object Detection, Video Analytics, Real-time Inference, NVIDIA TensorRT, Docker
- **Cloud & Infrastructure:** GCP (Workflows, Dataproc), AWS, Azure, Docker, Kubernetes, MLOps
- **Data & ML Platforms:** Apache Spark, Databricks, SQL, ETL, Data Validation

ACADEMIC QUALIFICATIONS

- **Postgraduate Degree in Data Science**
Pontifícia Universidade Católica do Paraná (Curitiba, Brazil)
- **Associate's Degree in Systems Analysis and Development**
Universidade Estácio de Sá (Curitiba, Brazil)
- **Web Development Certification**
Senai Paraná (Curitiba, Brazil)

PROFESSIONAL EXPERIENCE

Senior Computer Vision Engineer

Mediar Solutions – California, USA | April 2024 – Current

- Designed and deployed a production AI assistant using LLMs, tool calling, and agentic workflows to answer KPI questions in natural language directly from the company's database, served via FastAPI.
- Replaced a third-party SaaS detection system with a fully in-house Computer Vision pipeline across a multi-store retail environment (~500 cameras), eliminating significant recurring licensing costs and giving the business full ownership of its detection infrastructure.
- Fine-tuned YOLO models on real-world CCTV footage, achieving ~92% precision and ~88% recall at production confidence thresholds — validated across variable lighting, occlusion, and wide-angle retail camera feeds.
- Implemented Re-ID models to improve person tracking continuity across camera feeds, reducing identity-switching errors by ~45% compared to baseline detection alone.
- Migrated legacy object-detection workflows to Apache Spark on GCP Dataproc, cutting data processing time by ~40% and proportionally reducing cloud compute costs given GCP's time-based billing model.
- Optimized inference containers with Docker and NVIDIA GPU acceleration for production and edge-based VM deployment, making the pipeline viable for on-premise store infrastructure.
- Accelerated local training and prototyping workflows, reducing iteration cycles for behavioral vision models.
- Built automated ingestion pipelines on GCP Workflows to handle continuous video processing at scale, resolving previous reliability and timeout constraints.
- Architected a GDPR-compliant cloud environment for end-to-end Computer Vision workflows, implementing privacy-by-design controls across video ingestion, processing, storage, retention, and deletion, enabling the organization to meet key GDPR compliance requirements while supporting continued scale of multi-camera CV deployments without compliance risk.

Machine Learning Engineer

Medome – California, USA | Jun 2025 – Aug 2026

- Designed a multi-agent clinical extraction engine that transforms raw patient-provider dialogues into validated medical schemas, eliminating manual structuring and enabling reliable downstream diagnostic processing.
- Built the end-to-end RAG & knowledge pipeline for clinical data and documents, covering ingestion, processing, chunking, embeddings, vector storage, retrieval, context construction, and integration with AI agents.
- Designed the architecture to decouple knowledge ingestion and indexing from agent runtime, enabling both pipelines to scale and evolve independently while providing agents with relevant patient history, clinical documentation, and diagnostic knowledge.
- Built a real-time Medical Transcription & Scribe system combining STT and LLMs to automatically generate structured clinical notes and summaries during consultations, reducing documentation overhead for providers.
- Implemented an LLM Evaluation Framework with model traceability and HIPAA-ready privacy patterns, making AI outputs auditable in a regulated healthcare environment.
- Reduced LLM inference costs and latency through prompt engineering and structured output validation, keeping production operations lean without sacrificing output quality.
- Built an LLM Gateway providing a centralized abstraction layer across multiple LLM providers, allowing services to independently configure providers, models, parameters, and routing strategies.
- Eliminated redeployments for provider/model changes by moving LLM configuration from application code into service-level configuration, reducing engineering overhead and deployment risk across production services.

Senior Data Engineer

Grupo Marista – Curitiba, Brazil | June 2023 – April 2024

- Designed and maintained Python-based ML pipelines supporting predictive analytics for educational applications serving thousands of users.
- Built scalable ETL workflows (Spark, SQL) and migrated legacy systems to a modern data lake architecture, improving performance and data availability.

Senior Data Analyst

Meta IT – Brazil | June 2021 – May 2023

- Developed Python-driven data pipelines and SQL-based models powering BI applications consumed by large enterprise user bases.